Ethics Arena

Final Project Proposal: A Multi-Agent AI Debate Chatroom for Moral and Ethical Dilemmas

Prepared for the AI Tools final project. This proposal focuses on feasibility, architecture, page planning, and a realistic milestone path.

1. Overview

Ethics Arena is an interactive chatroom where multiple AI agents debate moral and ethical questions from different philosophical and social perspectives. Each agent acts as a distinct person with a stable identity, values, and speaking style. A human user can join the same discussion, challenge claims, ask follow-up questions, or request additional evidence.

The core idea is to move beyond a single-answer chatbot. Instead of producing one blended response, the system presents structured disagreement, sourced argumentation, and moderated discussion around difficult ethical issues.

2. Problem And Motivation

Ethical questions rarely have one universally accepted answer. Topics such as privacy, autonomous weapons, euthanasia, free speech, AI bias, and punishment are shaped by competing value systems. A standard chatbot often compresses disagreement into one summary, which hides the tension that makes these questions interesting.

This project addresses that gap by building a multi-agent debate environment that can represent several moral lenses at once while grounding claims in retrieved sources.

3. Objectives

- Build a chatroom-style product with multiple AI debate agents.
- Give each agent a stable role, ethical framework, and conversational style.
- Enable source-backed argumentation through external retrieval.
- Allow the user to actively intervene in the discussion.
- Generate clear summaries of agreements, disagreements, and unresolved tensions.

4. Core Features

Multi-Agent Debate

Example agents include a utilitarian thinker, a deontological thinker, a virtue ethics thinker, a pragmatic policy analyst, and a moderator.

Structured Turn Flow

Opening statements, evidence retrieval, rebuttals, user intervention, and a final moderator summary keep the debate organized.

Retrieval-Augmented Reasoning

Agents can draw on papers, analyses, public case studies, and selected media references instead of relying only on model memory.

User Participation

The user can question an agent, request evidence, challenge assumptions, or redirect the topic during the discussion.

5. Feasibility

Why This Is Realistic

- Multi-agent behavior can be orchestrated with one LLM backend plus strong role prompts.
- Search APIs can supply articles and paper metadata without building a crawler.
- A chatroom interface is straightforward with common web tooling.
- The main academic value is in orchestration, role design, evidence control, and interaction design.

Main Risks

- Agents may sound too similar if role prompts are weak.
- Low-quality sources can reduce argument quality.
- Without turn control, debates can become repetitive.
- Some ethics topics require moderation and safety limits.
- Latency and API cost increase as more agents and tools are added.

Scope Strategy

The project should begin with an MVP: three debate agents, one moderator, text-first chat, article or paper retrieval, structured turns, and source cards. Video and image support can be treated as stretch goals.

6. Technical Architecture

Frontend

- Next.js or React
- Chatroom layout
- Evidence card panel
- User input and control buttons

Backend

- Node.js or Next API routes
- OpenAI API for debate generation
- Search API integration for retrieval
- Session orchestration logic

State And Memory

- Current topic
- Agent roles and stance definitions
- Turn history
- Used citations and user interventions

Safety Layer

- Risky-topic detection
- Moderation rules
- Safer fallback responses
- Constraint checks on outputs

Suggested Stack

- Frontend: Next.js, React, Tailwind CSS or custom CSS
- Backend: Next API routes or Express
- AI: OpenAI API
- Retrieval: Tavily, SerpAPI, or Bing Search API
- Storage: SQLite during development
- Optional extensions: Semantic Scholar, Crossref, YouTube Data API

7. Page Plan

Home / Topic Setup

- Enter a debate topic
- Select debate mode or topic category

- Choose default agent roles
- Start a session

Debate Room

- Live multi-agent conversation timeline
- Participant list with role labels
- User input box
- Controls for summarize, rebut, or request evidence

Source Panel

- Title, source type, short summary, and link
- Visible connection between sources and agent claims

Summary View

- Strongest arguments by each side
- Agreements and disagreements
- Source list and final reflection

8. Development Milestones

1. Project setup and basic topic/chat UI.
2. Moderator-driven multi-agent debate engine.
3. Search and retrieval integration with evidence cards.
4. User participation and session memory.
5. Safety checks, final summaries, and UI polish.
6. Testing, screenshots, and final demo preparation.

9. Evaluation

The project can be evaluated by role distinctiveness, argument quality, evidence grounding, user participation quality, and conversation stability across multiple turns. A final class demo can show several ethics topics and compare debate quality with and without retrieval support.

10. Expected Outcome

The expected result is a working prototype that demonstrates multi-agent orchestration, retrieval-augmented reasoning, and an interactive debate experience around ethical dilemmas. A well-scoped MVP with strong role identity and reliable citation behavior would already make a compelling final project.

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